

## MATRIX ADDITION

```
#include<stdio.h>
#include<conio.h>
int main()
{
    int mat1[3][3], mat2[3][3], i, j, mat3[3][3];
    printf("Enter 3*3 matrix 1 elements :");
    for(i=0; i<3; i++)
    {
        for(j=0; j<3; j++)
            scanf("%d",&mat1[i][j]);
    }
    printf("Enter 3*3 matrix 2 elements :");
    for(i=0; i<3; i++)
    {
        for(j=0; j<3; j++)
            scanf("%d",&mat2[i][j]);
    }
    printf("\nAdding the two matrix.....");
    for(i=0; i<3; i++)
    {
        for(j=0; j<3; j++)
            mat3[i][j]=mat1[i][j]+mat2[i][j];
    }
    printf("\nBoth matrix added successfully!");
    printf("\nHere is the new matrix:\n");
    for(i=0; i<3; i++)
    {
        for(j=0; j<3; j++)
            printf("%d ",mat3[i][j]);
        printf("\n");
    }
    getch();
    return 0;
}
```

## MATRIX SUBTRACTION

```
#include <stdio.h>
#include <stdlib.h>

int main(void)
```

```

    {
        int r, c;
        printf("Enter the number of rows: ");
        scanf("%d", &r);
        printf("Enter the number of columns: ");
        scanf("%d", &c);
        int i, j;
        int a[r][c], b[r][c], res[r][c];
        printf("Enter the elements of first matrix:\n");
        for (i = 0; i < r; i++)
        {
            for (j = 0; j < c; j++)
            {
                printf("[%d][%d]: ", i, j);
                scanf("%d", &a[i][j]);
            }
        }
        printf("Enter the elements of second matrix:\n");
        for (i = 0; i < r; i++)
        {
            for (j = 0; j < c; j++)
            {
                printf("[%d][%d]: ", i, j);
                scanf("%d", &b[i][j]);
            }
        }
        for (i = 0; i < r; i++)
        {
            for (j = 0; j < c; j++)
            {
                res[i][j] = a[i][j] - b[i][j];
            }
        }
        printf("The result matrix is:\n");
        for (i = 0; i < r; i++)
        {
            for (j = 0; j < c; j++)
            {
                printf("%6d", res[i][j]);
            }
            printf("\n");
        }
    }
}

```

## MATRIX MULTIPLICATION

```
#include<stdio.h>
#include<stdlib.h>
int main(){
int a[10][10],b[10][10],mul[10][10],r,c,i,j,k;
printf("enter the number of row=");
scanf("%d",&r);
printf("enter the number of column=");
scanf("%d",&c);
printf("enter the first matrix element=\n");
for(i=0;i<r;i++)
{
for(j=0;j<c;j++)
{
scanf("%d",&a[i][j]);
}
}
printf("enter the second matrix element=\n");
for(i=0;i<r;i++)
{
for(j=0;j<c;j++)
{
scanf("%d",&b[i][j]);
}
}
printf("multiply of the matrix=\n");
for(i=0;i<r;i++)
{
for(j=0;j<c;j++)
{
mul[i][j]=0;
for(k=0;k<c;k++)
{
mul[i][j]+=a[i][k]*b[k][j];
}
}
```

```
}  
}  
//for printing result  
for(i=0;i<r;i++)  
{  
for(j=0;j<c;j++)  
{  
printf("%d\t",mul[i][j]);  
}  
printf("\n");  
}  
return 0;  
}
```